

Roll No.

TME-701

B. TECH. (ME/AME/Robotics) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

MECHANICAL VIBRATIONS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Determine the equations of the motion and natural frequency of the simple pendulum mass m , length l , spring of stiffness k attached at distance a , from hanging point. (Fig. 1). (CO2)

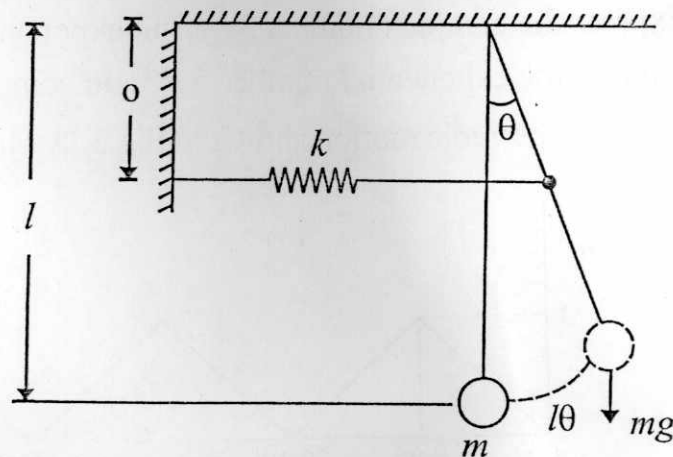


Fig. 1

P. T. O.

OR

- (b) Determine the equations of the motion and natural frequency of the pulley with mass M , mass moment of inertia I , radius r , attached with mass m at radius r and spring of stiffness k at radius r (Fig. 2). (CO1)

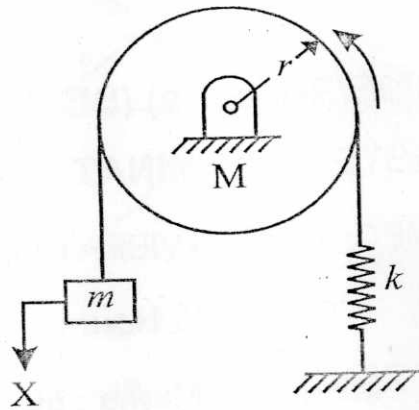


Fig. 2

2. (a) Add the following two motions analytically and check the solution graphically : (CO1)

- (i) $x_1 = 5 \sin (\omega t + 1)$
 (ii) $x_2 = 2 \cos (\omega t + 0.5)$

OR

- (b) (i) Express the complex number $3 + j 7$ in exponential form.
 (ii) Express the exponential number $5e^{j0.1}$ in complex form. (CO1)

3. (a) Represent the periodic motion shown in Fig 3 by Harmonic series. (CO1)

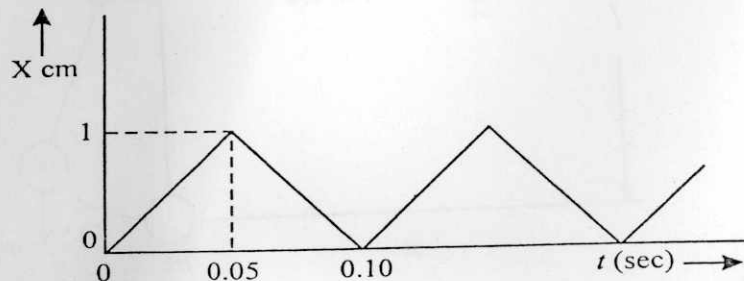


Fig. 3

(3)

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OR

(b) Represent the periodic motion shown in Fig. 4 by Harmonic series.

(CO1)

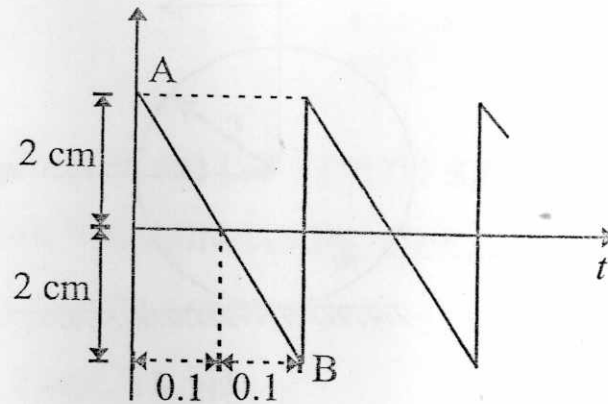


Fig. 4

4. (a) Give examples of the following by a neat sketch :

(CO1)

- (i) Longitudinal vibration
- (ii) Transverse vibration
- (iii) Torsional vibration

OR

(b) Define the following terms related to vibrations :

(CO1)

- (i) Cycle
- (ii) Period
- (iii) Frequency
- (iv) Degree of freedom
- (v) Simple Harmonic Motion

P. T. O.

5. (a) Determine the equations of the motion and natural frequency of the system shown in Fig. 5. (CO1, CO2)

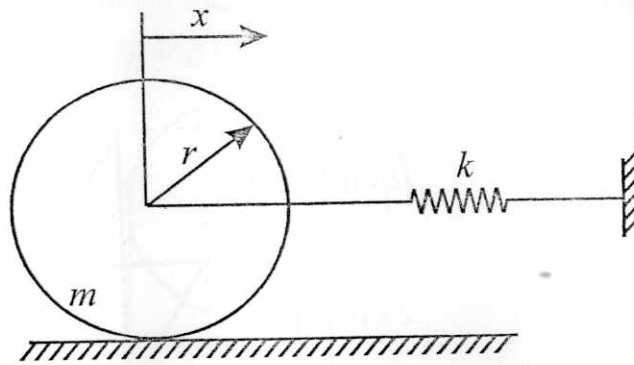


Fig. 5

OR

- (b) Add the following two motions analytically and check the solution graphically : (CO1, CO2)

$$x_1 = 4 \cos(\omega t + 10^\circ), \quad x_2 = 6 \sin(\omega t + 60^\circ)$$